ModelIt!

Student Activity Pack

Atoms Rearrange: Modeling Chemical Reactions

Conservation of Mass in Action

Grade 06L2

NGSS MS-PS1-2, MS-PS1-5

Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 When you burn a piece of paper, it seems to disappear. Where do you think the material goes?

2 If you mix baking soda and vinegar in a sealed bag, do you think the bag will weigh more, less, or the same after the reaction?

3 What do you think atoms do during a chemical reaction? Do they change into different atoms?

4 Draw what you think happens to atoms when wood burns.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Reactant Concentration | Temperature | Reaction Rate | Product Formation | Total Mass

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fast Reaction

High concentration, high temperature

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Slow Reaction

High concentration, low temperature

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Running Out

High start, observe decline over time

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Surface Area — Crushed or powdered reactants react faster because more molecules are exposed for collisions

Catalyst Presence — A substance that speeds up a reaction without being consumed, lowering the energy needed to start the reaction

Equilibrium Point — The state where forward and reverse reactions occur at equal rates, and concentrations stop changing

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If atoms are never created or destroyed in a chemical reaction, why do some reactions seem to make things disappear? Use evidence from your simulation to support your answers.

1. **Why do reactant and product lines on the graph look like mirror images of each other?**

2. **If mass is conserved, why does a log seem lighter after it burns?**

3. **Why does the reaction rate slow down even when temperature stays the same?**

4. **How is conservation of mass related to the idea that atoms are never created or destroyed?**

5. **How could you prove that a burned match has the same total mass as before, including all products?**

8 STEM Challenge: Design a Conservation of Mass Experiment

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YOUR ENGINEERING MISSION

Design and execute a sealed-system experiment that proves conservation of mass during a chemical reaction.

THE SCENARIO

A science magazine is challenging student teams to prove conservation of mass using everyday chemical reactions. Your team must design an experiment where you can measure the total mass before and after a reaction in a sealed system, proving that mass is conserved even when gases are produced.

Engineering Design Thinking

1. WHAT CHEMICAL REACTION WILL YOU USE (BAKING SODA + VINEGAR, STEEL WOOL RUSTING, ETC.)?
2. HOW WILL YOU SEAL THE SYSTEM SO NO GASES ESCAPE?
3. HOW WILL YOU MEASURE MASS ACCURATELY BEFORE AND AFTER THE REACTION?
4. WHAT EVIDENCE WILL SHOW THAT A CHEMICAL REACTION ACTUALLY OCCURRED (NOT JUST MIXING)?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Chemical Engineers design and optimize chemical reactions for manufacturing everything from medicines to fuels to plastics. They apply conservation of mass to scale reactions from lab to industrial

size. They earn $75,000-$140,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the chemical reaction model, what happens to the Total Mass line when reactants convert to products?

- **A.** Total Mass decreases because some matter is destroyed in the reaction
- **B.** Total Mass increases because new atoms are created during the reaction
- **C.** Total Mass stays perfectly flat because atoms are neither created nor destroyed, they just rearrange
- **D.** Total Mass fluctuates up and down as the reaction proceeds

2 A student burns a 200-gram log in an open fire and the remaining ash weighs only 15 grams. Does this violate conservation of mass?

- **A.** Yes, 185 grams of matter were destroyed by the fire
- **B.** No, the missing 185 grams escaped as carbon dioxide gas and water vapor into the atmosphere
- **C.** Yes, fire converts matter into pure energy
- **D.** No, the missing mass was absorbed by the surrounding air

3 Why does the reaction rate slow down as a chemical reaction proceeds, even if the temperature stays constant?

- **A.** The products interfere with the reaction and block it from continuing
- **B.** The container walls absorb energy from the reaction
- **C.** As reactant molecules are consumed, fewer molecules remain to collide, so collisions become less frequent
- **D.** Chemical reactions naturally lose energy over time

4 How does increasing temperature affect a chemical reaction?

- **A.** Higher temperature creates new atoms that speed up the reaction
- **B.** Higher temperature makes molecules move faster, collide more often, and collide with more energy, increasing the reaction rate
- **C.** Higher temperature has no effect on reaction rate, only on the amount of product formed
- **D.** Higher temperature destroys some reactant molecules, slowing the reaction

5 In the model, reactant concentration and product formation are described as 'mirror images' of each other. What does this mean and why does it happen?

- **A.** Reactants and products are always at the same concentration
- **B.** As reactant concentration decreases by a certain amount, product formation increases by exactly the same amount, because atoms are rearranging rather than being created or destroyed
- **C.** The two graphs look the same but are measured at different times
- **D.** Products break down into reactants at the same rate they are formed